

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **Methyl dichloroacetate**  
 Product Description: **Methyl dichloroacetate**  
 Cat No. : 148620000; 148621000; 148625000  
 CAS No 116-54-1  
 Molecular Formula C3 H4 Cl2 O2

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

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 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                   |
|----------------------------------------------------|-------------------|
| Skin Corrosion/Irritation                          | Category 2 (H315) |
| Serious Eye Damage/Eye Irritation                  | Category 2 (H319) |
| Specific target organ toxicity - (single exposure) | Category 3 (H335) |

**Label Elements**

**Signal Word**
**Warning**
**Hazard Statements**

H335 - May cause respiratory irritation

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H319 - Causes serious eye irritation  
H315 - Causes skin irritation

## Precautionary Statements

### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P264 - Wash face, hands and any exposed skin thoroughly after handling

### Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Combustible liquid

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component              | CAS No   | Weight % |
|------------------------|----------|----------|
| Methyl dichloroacetate | 116-54-1 | 99       |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention.

#### Ingestion

Clean mouth with water. Get medical attention.

#### Inhalation

Remove from exposure, lie down. Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Get medical attention.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

Difficulty in breathing. . Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

### Indication of any immediate medical attention and special treatment needed

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## Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Combustible material. Flammable. Containers may explode when heated.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen chloride gas.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental precautions

See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Avoid contact with skin and eyes. Do not breathe mist/vapors/spray. Keep away from open flames, hot surfaces and sources of ignition.

### Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

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## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

#### Exposure Controls

##### **Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

##### **Personal protective equipment**

###### **Eye Protection**

Goggles

###### **Hand Protection**

Protective gloves

###### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

###### **Respiratory Protection**

No protective equipment is needed under normal use conditions

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### **Appearance**

Colorless

#### **Physical State**

Liquid

#### **Odor**

Petroleum distillates

#### **Odor Threshold**

No data available

#### **pH**

No information available

#### **Melting Point/Range**

-52 °C / -61.6 °F

#### **Softening Point**

No data available

#### **Boiling Point/Range**

141 - 145 °C / 285.8 - 293 °F

@ 760 mmHg

#### **Flash Point**

80 °C / 176 °F

**Method -** No information available

#### **Evaporation Rate**

No data available

#### **Flammability (solid,gas)**

Not applicable

Liquid

#### **Explosion Limits**

No data available

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|                              |                          |             |
|------------------------------|--------------------------|-------------|
| Vapor Pressure               | 4.7 hPa @ 20 °C          |             |
| Vapor Density                | 4.93                     | (Air = 1.0) |
| Specific Gravity / Density   | 1.381                    |             |
| Bulk Density                 | Not applicable           | Liquid      |
| Water Solubility             | Slightly soluble         |             |
| Solubility in other solvents | No information available |             |

## Partition Coefficient (n-octanol/water)

|                           |                          |                                        |
|---------------------------|--------------------------|----------------------------------------|
| Autoignition Temperature  | No data available        |                                        |
| Decomposition Temperature | No data available        |                                        |
| Viscosity                 | No data available        |                                        |
| Explosive Properties      |                          | explosive air/vapour mixtures possible |
| Oxidizing Properties      | No information available |                                        |

|                   |              |
|-------------------|--------------|
| Molecular Formula | C3 H4 Cl2 O2 |
| Molecular Weight  | 142.97       |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
| Hazardous Reactions      | No information available. |

### Conditions to Avoid

Keep away from open flames, hot surfaces and sources of ignition.

### Incompatible Materials

Acids. Bases. Reducing Agent.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

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|                                                   |                                                                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product Information</b>                        | No acute toxicity information is available for this product                                                                         |
| <b>(a) acute toxicity;</b>                        |                                                                                                                                     |
| Oral                                              | No data available                                                                                                                   |
| Dermal                                            | No data available                                                                                                                   |
| Inhalation                                        | No data available                                                                                                                   |
| <b>(b) skin corrosion/irritation;</b>             | Category 2                                                                                                                          |
| <b>(c) serious eye damage/irritation;</b>         | Category 2                                                                                                                          |
| <b>(d) respiratory or skin sensitization;</b>     |                                                                                                                                     |
| Respiratory                                       | No data available                                                                                                                   |
| Skin                                              | No data available                                                                                                                   |
| <b>(e) germ cell mutagenicity;</b>                | No data available                                                                                                                   |
| <b>(f) carcinogenicity;</b>                       | No data available                                                                                                                   |
|                                                   | There are no known carcinogenic chemicals in this product                                                                           |
| <b>(g) reproductive toxicity;</b>                 | No data available                                                                                                                   |
| <b>(h) STOT-single exposure;</b>                  | Category 3                                                                                                                          |
| Results / Target organs                           | Respiratory system.                                                                                                                 |
| <b>(i) STOT-repeated exposure;</b>                | No data available                                                                                                                   |
| Target Organs                                     | No information available.                                                                                                           |
| <b>(j) aspiration hazard;</b>                     | No data available                                                                                                                   |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated. See actual entry in RTECS for complete information                   |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.                                                |
| <b>Endocrine Disrupting Properties</b>            | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |

## SECTION 12: ECOLOGICAL INFORMATION

|                                             |                                                                                                                             |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b><u>Ecotoxicity effects</u></b>           | Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. |
| <b><u>Persistence and degradability</u></b> |                                                                                                                             |
| Persistence                                 | Soluble in water, Persistence is unlikely, based on information available.                                                  |

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|                                               |                                                                                                                                                              |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Bioaccumulative potential</u></b>       | Bioaccumulation is unlikely                                                                                                                                  |
| <b><u>Mobility in soil</u></b>                | The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils. |
| <b><u>Endocrine Disruptor Information</u></b> | This product does not contain any known or suspected endocrine disruptors                                                                                    |
| <b><u>Other adverse effects</u></b>           | No information available                                                                                                                                     |

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                                   |                                                                                                                                                                     |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Waste treatment methods</u></b>             |                                                                                                                                                                     |
| <b><u>Waste from Residues/Unused Products</u></b> | Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations |
| <b><u>Contaminated Packaging</u></b>              | Dispose of this container to hazardous or special waste collection point.                                                                                           |
| <b><u>Other Information</u></b>                   | Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains                                         |

## SECTION 14: TRANSPORT INFORMATION

|                        |                        |
|------------------------|------------------------|
| <b><u>IMDG/IMO</u></b> |                        |
| UN-No                  | UN2299                 |
| Hazard Class           | 6.1                    |
| Packing Group          | III                    |
| Proper Shipping Name   | METHYL DICHLOROACETATE |

|                                       |                        |
|---------------------------------------|------------------------|
| <b><u>Road and Rail Transport</u></b> |                        |
| UN-No                                 | UN2299                 |
| Hazard Class                          | 6.1                    |
| Packing Group                         | III                    |
| Proper Shipping Name                  | METHYL DICHLOROACETATE |

|                      |                        |
|----------------------|------------------------|
| <b><u>IATA</u></b>   |                        |
| UN-No                | UN2299                 |
| Hazard Class         | 6.1                    |
| Packing Group        | III                    |
| Proper Shipping Name | METHYL DICHLOROACETATE |

|                                            |                                 |
|--------------------------------------------|---------------------------------|
| <b><u>Special Precautions for User</u></b> | No special precautions required |
|--------------------------------------------|---------------------------------|

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                                         |            |
|-----------------------------------------|------------|
| <b><u>International Inventories</u></b> | X = listed |
|-----------------------------------------|------------|

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| Component              | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Methyl dichloroacetate | 204-146-8 | X    | -   | X     | X    | X    | X     | -    | KE-23755 |

## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date**

21-Mar-2025

**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**